

Package level thermal analysis of backside power delivery network (BS-PDN) configurations

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Abstract— This paper presents a thermal modeling analysis of the impact of the introduction of a backside power delivery network (BS-PDN) and assesses the trade-off between the beneficial and detrimental thermal consequences of this BS-PDN. In the BS-PDN concept, the power delivery network is moved to the backside of the chip, which enables enhanced system performance, increased chip-area utilization and reduced fine metal pitch BEOL complexity. The connectivity between the backside metals of the PDN and the transistors is provided by means of micro or nano through-Si vias, which requires an extreme thinning of the Si substrate. The potential self-heating issue due to this substrate thinning is the main thermal concern for the backside PDN concept. This paper focuses on the thermal modeling study for a *TSV-last BS-PDN concept*, for which the TSVs through the Si substrate are processed in the last step, after the processing of the front-end and power rails. A unit cell thermal finite element model of the via-last BS-PDN test case is used with size dependent thermal properties, derived from dedicated nanoscale Monte Carlo simulations. The simulation studies assess the thermal impact of the Si substrate thickness, the backside metals, and the density of the μ TSVs.

Keywords—3D stacking, 3D-IC, thermal modeling

Nomenclature

BEOL	Back-end of line
BPR	Buried power rail
BS	Back side
BSPR	Backside power rails
FEOL	Front-end of line
FS	Front side
M1	Backside metal 1 layer
P	Pitch
PCB	Printed circuit board
PDN	Power delivery network
RDL	Re-distribution layer
STI	Shallow trench isolation
T	Temperature, (°C)
TSV	Through-silicon via

I. INTRODUCTION

The goal of a power delivery network (PDN) is to provide power and reference voltage to the active devices on the die, in the most efficient way. This PDN is essentially a network of interconnects that is separate from the signal network. Traditionally, both networks are fabricated through back-end-of-line (BEOL) processing on the frontside of the wafer. The power delivery network thus competes for space with the signal network. This results in a congestion of the routing of the signals for scales transistor nodes. However, it is also possible to move the power distribution to the backside of the silicon wafer, which today serves only as a carrier. This would allow direct power delivery to the standard cells, and promises to enhance system performance, increase chip area utilization, improve power integrity, and reduce the fine metal pitch BEOL complexity. This concept is referred to as a *backside power delivery network* [1-4] (BS-PDN) and combines 3D – through silicon vias (TSVs) technology and logic technology to decouple the power grid from the design budget and deliver power from the backside of a thinned device wafer. Fig. 1 shows a schematic comparison of the conventional frontside PDN and the proposed back-side power delivery configuration. The TSVs, that connect the backside metal to the BPR, are referred to as nanoTSVs or μ TSVs, have a size in the order of 100 nm. Since the aspect ratio of the TSVs is limited for etching and filling, the Si substrate needs to be thinned.

Backside power delivery configurations are typically categorized based on the processing steps order, schematically shown in Fig. 2. For the TSV-last option, as the name suggests, the processing of the TSVs is performed last through the thinned Si substrate, after the processing of the FEOL and the buried power rails (BPR). In the case of the TSV-middle option, the TSVs are processing between the FEOL, and the backside power rails (BSPR). For the TSV-last BSPDN configuration, the typical Si substrate is several hundreds of nm, and even thinner for the TSV-middle concepts (several tens of nm or even less).

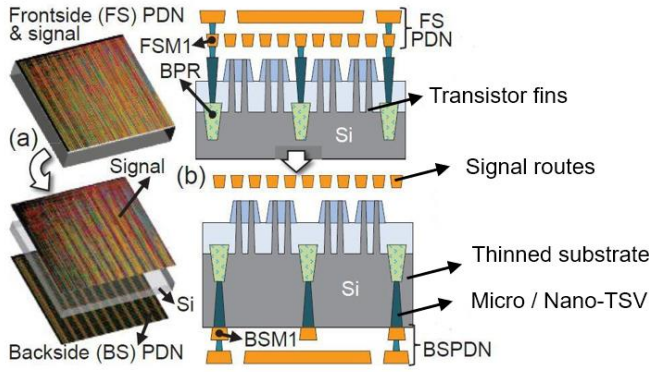


Fig. 1. Schematic (a) and cross-section (b) of conventional frontside (FS) power delivery configuration and backside (BS) metals for PDN and signal routing for the TSV-last concept. [3].

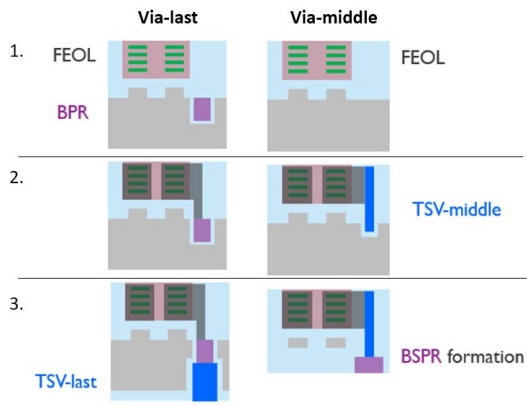


Fig. 2. Schematic comparison of the via-last BS-PDN integration approach (left) with buried power rails (BPR) and the via-middle BS-PDN integration approach (right) with backside power rails (BSPR).

One of the concerns for the introduction of backside power delivery configuration, is the thermal impact on the device self-heating due to the extreme thinning of the Si substrate. In literature, not many studies are available on this subject. To the best of the authors' knowledge, the only study on the thermal performance of the backside PDN, was reported by Hossein et al. [5]. They present a thermal modeling study that shows that the backside PDN configuration has a negligible impact on the temperature distribution of the die. However, in their simplified analysis, they only consider the impact of the additional thermal resistance for one-dimensional vertical heat transfer of the oxide bonding layer between the thinned die and the carrier compared to the thermal resistance of the air-cooled heat sink. A more detailed analysis is required to study the impact of both the additional layers as well as the extreme substrate thinning on the device self-heating.

This paper presents a modeling analysis of the thermal impact of the introduction of the backside power delivery network and assesses the trade-off between the beneficial and detrimental thermal consequences of the via-last BS-PDN. In Section II, the different local and global level thermal aspects of the BS-PDN configuration and wafer thinning are introduced. Section III presents the via-last PS-PDN test case

for the modeling study and the modeling approach and assumptions. In Section IV, the modeling results for the thermal comparison between the conventional FS-PDN and the BS-PDN configuration and the parameter sensitivity are discussed.

II. BS-PDN THERMAL CONSIDERATIONS

In order to assess the thermal performance comparison of the conventional frontside and the backside power delivery configurations, both the local heat spreading impact as well as the package level thermal behavior should be considered. The local level will mainly impact the thermal behavior of hot spots and locally distributed power dissipation, while the package level mainly impacts the average power. Since the power dissipation pattern in a chip is a combination of the uniform background power and localized hot spots, both the local and the package level contribute to the thermal impact of the BS-PDN.

On the *local* level, the main thermal drawback of the BS-PDN, is the reduction of the Si substrate thickness to a few hundred nm. For via-last BS-PDN, the Si substrate is typically in the order of 300 – 500 nm, while for other BS-PDN concepts, such as via-middle, the substrate can be even significantly thinner, down to several tens of nm. This drastic substrate thinning results in a significant reduction of the lateral spreading in the substrate and a reduction of the effective thermal conductivity, and thus consequently in a corresponding temperature increase compared to the self-heating in a thick substrate. This increase of the self-heating temperature for thinned substrates has been numerically [6] and experimentally demonstrated for devices [7] and thermal test structures [8-10]. The impact of the substrate thickness can be estimated using the thermal spreading / constriction resistance [11], as function of the heat sources size and the boundary condition at the bottom of the substrate. Fig. 3 shows the impact of the substrate thickness on the thermal resistance for different heat source sizes. This graph clearly shows a significant increase for reducing substrate thickness. At the other hand, the BS-PDN configuration introduces several backside metal layers (typically 3-4 layers). The presence of these relatively wide backside metal layers with a high metal density can help to spread the heat laterally, which has a beneficial impact on the device temperature and can help to compensate the negative impact of the substrate thinning.

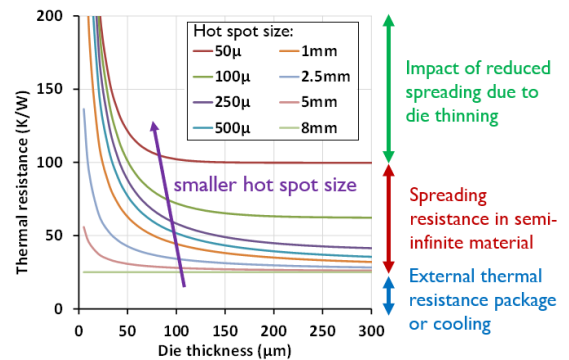
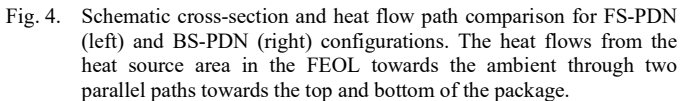


Fig. 3. Thermal resistance impact of substrate thinning for different heat source sizes on a 8x8 mm² Si substrate ($R_{ext} = 20$ K/W).

In the case of a *high-performance computing* application (HPC), the upward heat path towards the topside is dominant, where an efficient cooling solution (air cooling or liquid cooling) is provided. Typically, > 90% of the generated power is removed through the topside of the package [12]. Compared to the FS-PDN configuration, the heat needs to flow through the additional thermal resistances of the M_x and M_y BEOL layers and through the oxide bonding layer, resulting in an additional temperature increase. For *mobile applications*, where the package is used in a smartphone or tablet, both the upward and the downward path are significant, while their actual contribution depends on the connection of the package and PCB to the device enclosure through gap fillers or heat spreaders [13-17]. On the package level, the different materials in the thermal paths]. In case of the BS-PDN configuration, the heat flowing down from the FEOL towards the PCB faces a lower thermal resistance since the M_x and M_y are no longer in that thermal path, while the upward thermal path has a higher thermal resistance. The overall package level thermal impact of the introduction of the BS-PDN will therefore depend on the application and the assumed boundary conditions.



A. Via last BS-PDN test case

[illegible]

Table 1. Test case dimensions

Backside metal M1 pitch	500 nm
Backside metal M1 pitch	250 nm
Backside metal M1 height	210 nm
Buried power rail pitch	180 nm
Buried power rail width	30 nm
μTSV	180 nm x 250 nm
Large heat source	2500 nm x 1000 nm
Small heat source	400 nm x 1000 nm

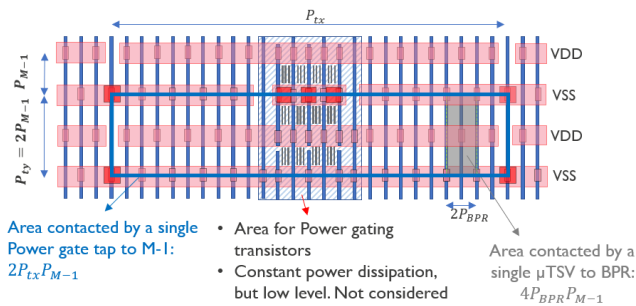


Fig. 6. Illustration of the small heat source and large heat source area on the grid on backside metal lines and buried power rails.

For the modeling study, a 3D stacked IC package is considered assuming boundary conditions representing a high-performance heat sink on top of the package. A 500 nm thick Si substrate is assumed, while a single backside metal layer is included. A 3D FE unit cell model of $200 \times 200 \times 300 \mu\text{m}^3$ is considered based on the pitch of the V_{dd} and V_{ss} bump array, as shown in Fig. 7a. A detail of the BPRs and backside metal lines is shown in Fig. 7b.

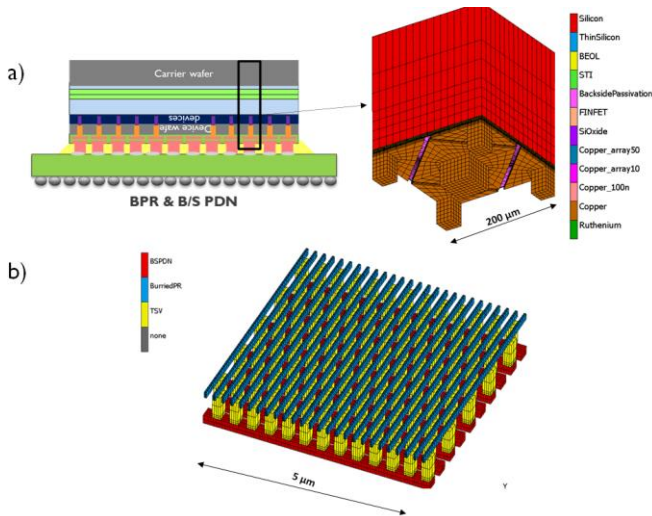


Fig. 7. 3D finite element thermal model: a) package level unit cell for the power delivery, b) $5\text{ }\mu\text{m} \times 5\text{ }\mu\text{m}$ model detail of backside metal lines, TSVs and buried power rails.

B. Thermal modeling approach

The feature sizes considered in the current analysis (<100 nm) are beyond the capabilities of continuum modeling methods and require nano-scale thermal modeling approaches for an accurate prediction of the thermal behavior. The effective thermal conductivity reduces at small dimensions due to relative increase of heat carriers' scattering at interfaces and within material. Since the objective of this is a relative thermal comparison between the FS-PDN and BS-PDN configuration considering the package level boundary conditions, rather than a very accurate detailed nano-scale simulation, a simplified modeling approach has been used. A Fourier heat conduction model has been combined with the use of effective size dependent material properties, calculated by dedicated Monte Carlo BTE simulations. For the dielectric material in the backside power delivery network, SiO₂ is considered with a

thermal conductivity of 1.4 W/m-K. The effective thermal conductivity values of the materials (Si substrate, Cu for the backside metal lines, Ru for the buried power rails, W for the μ TSVs) have been extracted from Monte Carlo BTE simulations as function of the dimensions of the relevant structures. 3D Monte Carlo BTE simulations with ab-initio phonon properties were performed using a custom-built code for the 3D geometry shown in Fig. 8. At the ‘inlet’ side, a uniform power boundary condition is applied, while at the ‘outlet’ side, an isothermal boundary condition is applied. In the simulations, diffuse surfaces (worst case for boundary scattering) and no grain boundaries (best case for intrinsic scattering) have been assumed. The effective thermal conductivity of the material under test is extracted from the linearized temperature difference over the considered test section L_{dut} , which excludes the inlet and outlet portions of the structures, in order to avoid the impact of the boundary conditions.

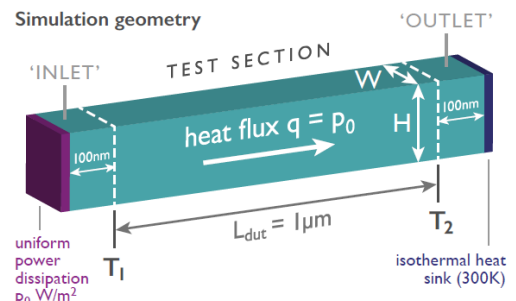


Fig. 8. Monte Carlo BTE simulation geometry for the calculation of the effective thermal conductivity values as function of the dimensions.

These effective values as function of the layer thickness for Si or the metal width are shown in Fig. 9. It can be seen that the thermal conductivity reduction for metal starts at lower dimensions, and exhibits a sharper transition compared to Si. This is because the metals can be well approximated with a single mean free path (29 nm for Cu, 7 nm for Ru) while there is a wide range of phonon mean free path values for Si (10 nm – 10 μ m). At the considered test case dimensions, the Si thermal conductivity reduces by almost a factor 2, which has an additional impact of 50% on the temperature increase, on top of the thermal spreading resistance due to the substrate thinning, shown in Fig. 10.

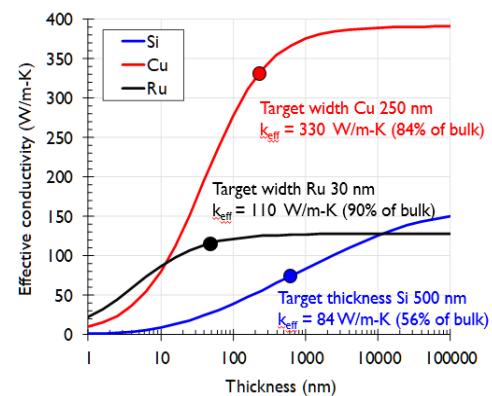


Fig. 9. Size dependent effective thermal conductivity values for Cu, Ru and Si, extracted from nanoscale Monte Carlo simulations.

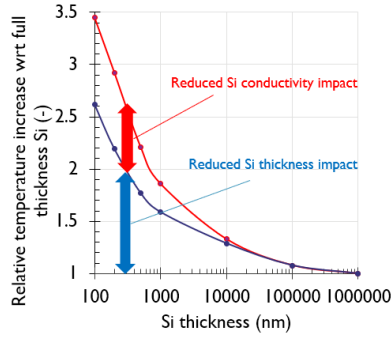


Fig. 10. Normalized temperature effect for the combined impact of the Si substrate thinning and Si thermal conductivity reduction.

IV. RESULTS

A. Thermal impact of BS-PDN

Fig. 11 shows the temperature distribution in the STI layer where the heat generation is modeled for both the small and large heat source, assuming of dense array of μ TSVs. The impact of the Si substrate thickness is shown in Fig. 12 for the case with and without backside metal. The thermal simulations show that the Si thickness reduction from the reference case with 200 μm to 500nm causes a 2.2X temperature increase in the case without backside metals. The presence of the single backside metal layer, with a dense array of μ TSVs results in a temperature reduction of 27% (small heat source) and 30% (large heat source), shown in Fig. 13. This results in an overall temperature increase of 60% for the BS-PDN case compared to the conventional FS-PDN configuration. This 60% temperature increase corresponds to a temperature increase for a thinned Si substrate thickness of 2.5 μm without backside metals. Therefore, the use of the backside can partially compensate for the reduced thermal spreading in the substrate and thus enable further Si thinning. The additional lateral thermal spreading is clearly visible in the heat flux contour plot and vector plot in Fig. 14, where the preferential lateral spreading in the metal lines can be observed. Also, the injection of heat into the metal lines at the locations of the μ TSVs is visible. The BS metal lines provide additional thermal spreading, to partially compensate the reduced spreading in the thinned Si and thus enable further Si thinning for more aggressive BS-PDN dimensions from a thermal point of view. Furthermore, the BS- μ TSVs (openings in BS passivation) are instrumental to get the heat into the BS metal, as is shown by the sensitivity study for the density of the TSVs.

Compared to the negligible thermal impact of BS-PDN, reported by Hossen et al. [5] for uniform power assumptions with 1-D vertical heat flow, the results in the present work show that there is a significant impact of the BS-PDN configuration for small heater sources: for a reduction of the Si substrate down to 500 nm, a temperature increase of 60% is observed (120%) in the absence of backside metals. Both the uniform heating case and the single hot spot case are extreme cases for the analysis of the thermal impact of the backside PDN. How close the impact for a certain application will be to either of the two extremes, depends on the actual power distribution map, and the presence of hot spots for the considered application.

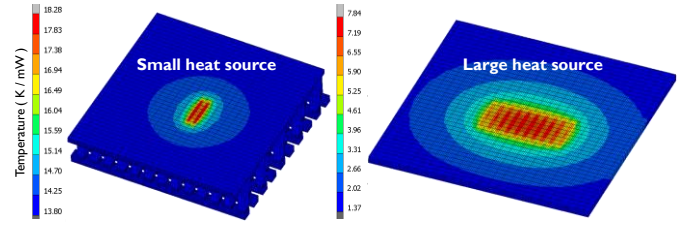


Fig. 11. Temperature contour plots for the small and large heat source test cases with backside metals for a heater area power of 1mW.

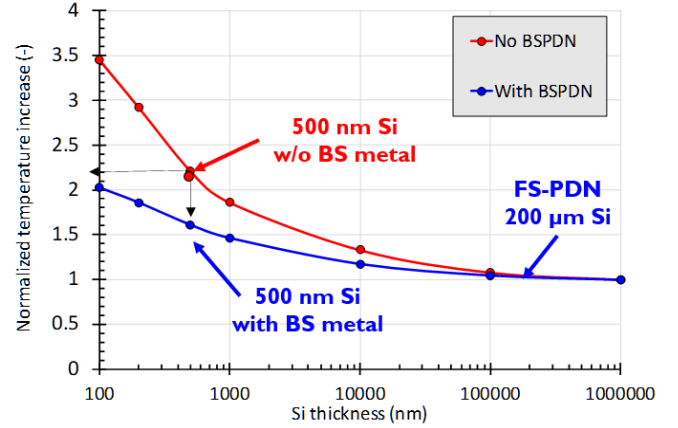


Fig. 12. Normalized temperature increase for the self-heating in case of thinned Si substrate with and without the presence of backside metal.

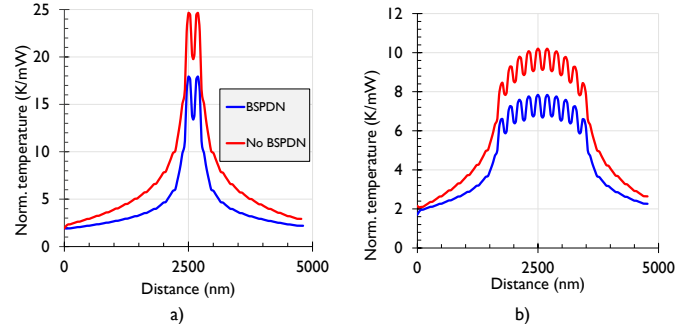


Fig. 13. Temperature profiles for the small (a) and large (b) heat sources with and without the presence of the backside metals.

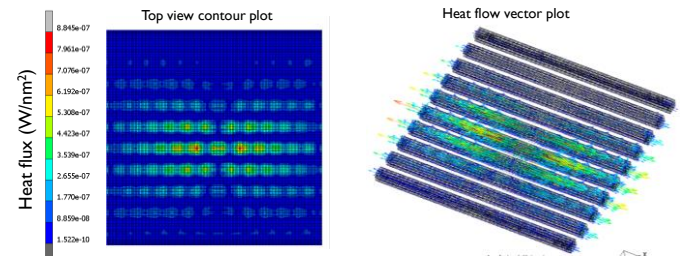


Fig. 14. Contour plot (left) and vector plot (right) of the heat flux in the backside oxide / metal layer.

B. Parameter sensitivity study: μ TSV density impact

The previous results showed that the thermal connection between the BPRs and the backside metal, provided by the μ TSVs are important to get the heat into the backside metal lines, where it can spread out laterally. For the analysis in the

previous section, a high μ TSV density of 16.7% was considered. However, for more realistic implementations, such as the tap cell implementation [19], lower μ TSV densities will be used. An example for the μ TSV density for such an implementation is 4%. In Fig. 15, the thermal behavior is compared for the large heat source case for a 500 nm thick Si substrate: 1) without BS metals; 2) with 1 BS metal layer and 4% μ TSV density; 3) with 1 BS metal layer and 16.7% μ TSV density. The average heat source temperature for the case with 4% μ TSV density is 12% lower compared to the case with backside metals and μ TSVs. This corresponds to an increase of 1.9x with respect to the reference case with 200 μ m Si substrate thickness, compared to 1.6x for the 16.7% μ TSV density, and 2.2x for the case with backside metal.

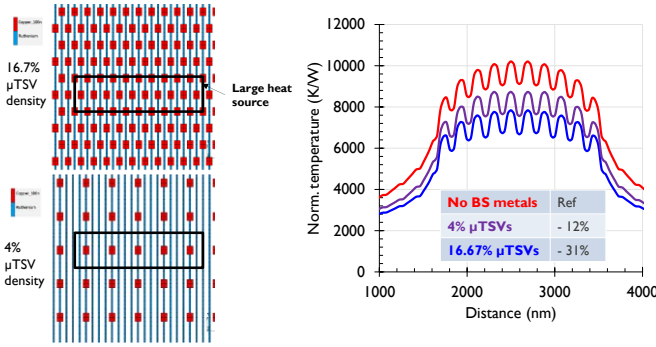


Fig. 15. Thermal impact of μ TSV density (no TSVs, 4% density, 16.7% density) on the temperature of the large heat source.

C. Sensitivity study: Backside metal thickness and pitch

Another parameter that can be varied is the thickness of the backside metal lines. In the reference simulation, a value of 210 nm was used. For the parameter sensitivity study, this value was varied in the range from 100 nm to 700 nm for a pitch of 500 nm. Fig. 16a shows the normalized temperature increase as function of the Si substrate thickness for the different backside metal thickness values. In Fig. 16b, the relation between the Si substrate thickness and the metal thickness in order to keep a constant relative temperature increase of 1.6x compared to the 200 μ m thick substrate without backside metals. When the Si substrate thickness is reduced from 500 nm to 300 nm, the temperature can be kept constant by increase the metal thickness to 700 nm. This indicates that the thermal impact of Si thickness reduction can be partially compensated by the backside metals.

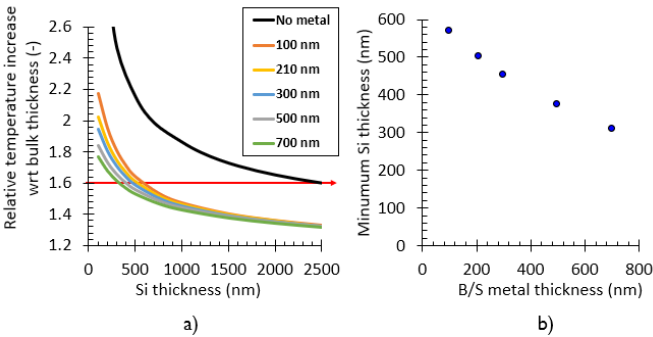


Fig. 16. Thermal impact of the Si substrate and BS metal thickness.

D. Transient analysis

The power generation in the transistors is an inherently dynamic process, where only part of the transistors is active during a short time, while the others are switched off. To evaluate the dynamic impact of the presence of the BS-PDN, a transient thermal simulation has been performed using the material properties shown in Table 2. The volumetric thermal capacitance of the Cu is 90% higher than the one of oxide, resulting in a slower heating and a lower delay. This thermal delay is visible from 10^{-9} s onwards in Fig. 17 where the transient response is compared for the case with and without backside metals.

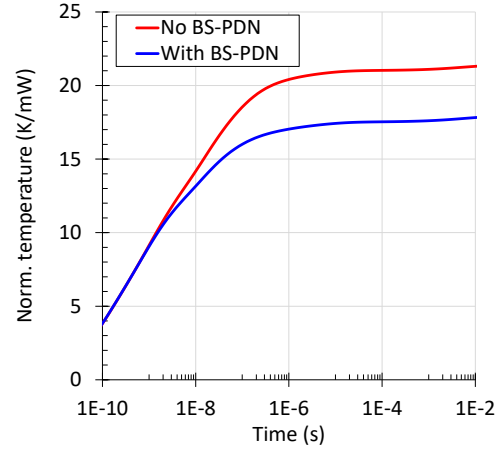


Fig. 17. Transient thermal impact of the presence of the backside metals.

Table 2. Material properties of the backside materials

Material	Density kg/m ³	Specific heat J/kg-K	Thermal capacitance kJ/m ³ -K
Cu	8940	385	3442
SiO ₂	2650	680	1802

V. CONCLUSIONS

This paper presents a thermal modeling analysis of the impact of the introduction of the backside power delivery network and assesses the trade-off between the beneficial and detrimental thermal consequences of this BS-PDN. The main thermal drawback is the reduction of the Si substrate thickness to a few hundred nm which results in a significant reduction of the lateral spreading in the substrate and a reduction of the effective thermal conductivity, and thus consequently in a corresponding temperature increase. The advantages from thermal point of view are the reduction of the remaining front-side BEOL stack (lower thermal resistance) and the presence of the backside metal layers with a high metal density which can help to spread the heat laterally. A unit cell thermal finite element model of a via-last BS-PDN test case has been built with size dependent thermal properties, derived from dedicated nanoscale Monte Carlo simulations.

The thermal simulations show that the Si thickness reduction from 200 μ m to 500 nm causes a 2.2x temperature increase for $2.5 \times 1 \mu\text{m}^2$ and $0.4 \times 1 \mu\text{m}^2$ heat sources without the presence of backside metals. The presence of a single backside metal layer with 16.7% μ TSV density results in a temperature

reduction of 27-30% for the considered heat source area. This corresponds to a thermal penalty of 60% for the TSV-last BS-PDN configuration with 500 nm Si substrate thickness compared to the conventional FS-PDN configuration. The backside metal lines provide additional thermal spreading, to partially compensate the reduced spreading in the thinned Si and thus enable further Si thinning for more aggressive BS-PDN dimensions from a thermal point of view. Furthermore, the BS- μ TSVs (openings in BS passivation) are instrumental to get the heat into the BS metal, as is shown by the sensitivity study for the density of the TSVs. When the μ TSV is reduced from 16.7% to 4%, the relative temperature increase of the BS-PDN increases from 1.6 to 1.9. Finally, the presence of the backside metals, close to the heat sources results in an increase of the thermal capacitance of the structures and helps to reduce the temperature peaks in dynamic operation.

The simulation results clearly show that the assumptions for the power distribution have a very large impact on the predicted effect of the introduction of the backside PDN. For a uniform power assumption, negligible thermal impact is reported, which depends on the vertical 1-D heat flow through the additional materials in the cross-section and associated thermal resistance. For a single hot spot assumption, the impact of the BS-PDN can be as high as a factor 2, dominated by the thermal impact of the substrate thinning. This assumption might however be too pessimistic. This large difference shows the need for a detailed thermal simulation, taking into account the power density distribution and the size of the considered hot spots for an accurate analysis of the thermal impact of the power delivery concepts.

VI. REFERENCES

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